

P2 - Analytics Engineering Platform

Portfolio Project 2 | dbt + Airflow + Docker | RC2

Business scenario

A subscription company has raw billing and customer data but inconsistent analytics logic. Build a containerized analytics engineering platform with modeled staging/intermediate/marts, dbt tests/documentation, Airflow orchestration, safe incremental behavior and reproducible local execution.

Required deliverables

- Define facts/dimensions and grain before modeling.
- Build dbt sources, staging, intermediate and marts using ref().
- Use generic and custom tests for uniqueness, relationships and business rules.
- Use an incremental model and prove rerun safety.
- Orchestrate source load -> dbt build/test -> publish in Airflow.
- Package reproducible services/config in Docker Compose without plaintext secrets.
- Handle late input and backfill/reprocess safely.
- Trigger and repair a supplied dbt uniqueness failure and a simulated Airflow partial failure.
- Deliver lineage/README/runbook and technical defense.

Deliberate failure pack

- Duplicate business key causing uniqueness test failure
- Broken relationship
- Late-arriving billing file
- Incremental duplicate risk
- Broken dbt ref
- Airflow upstream failure
- Downstream must not publish after failed quality test

Submission rule

Run the local project checker, preserve failure/recovery evidence, and save a final README + architecture + runbook. Live tool/runtime requirements remain conditional where the build environment cannot execute them.